AI Model with Road Network Planning

1. Course Content

- This course is a comprehensive advanced case study that combines AI large model functionality on the basis of road network planning navigation to achieve AI-controlled navigation within the road network

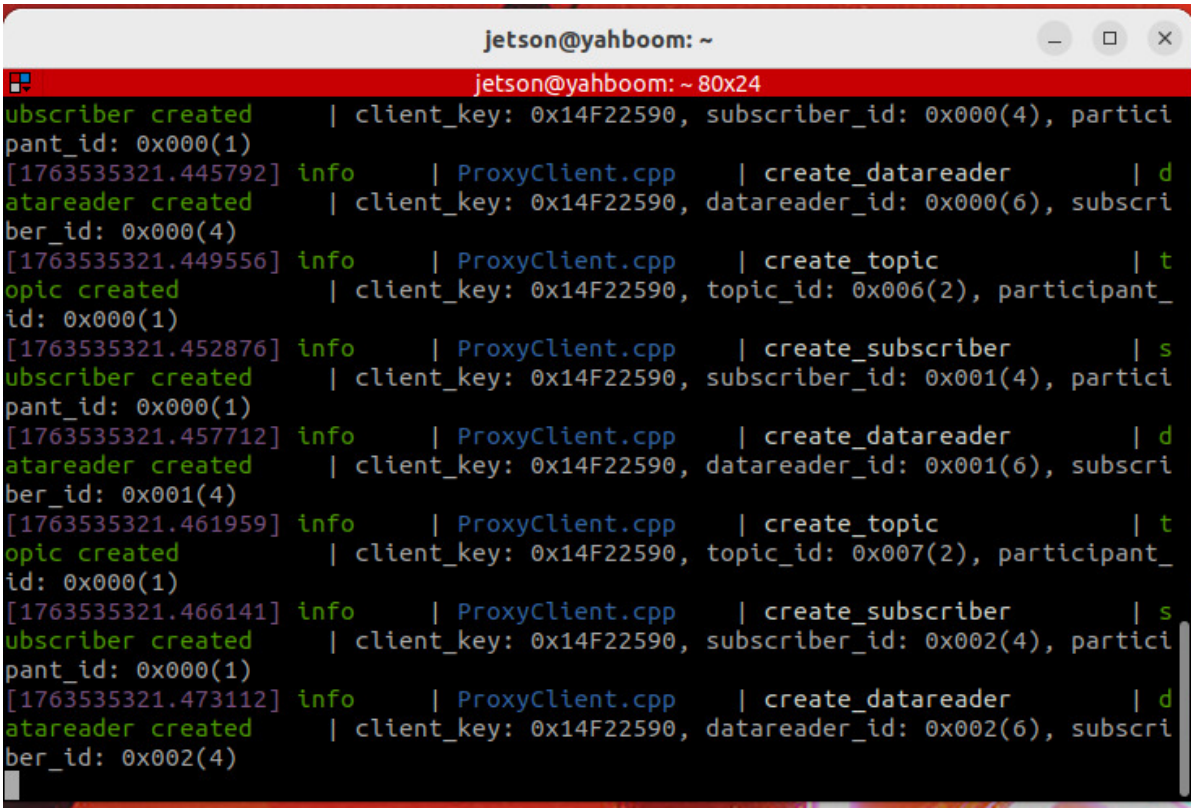
2. Preparation Work

2.1 Start `kilted_agent` Agent

- Use the following command to start the microROS agent and connect to the chassis

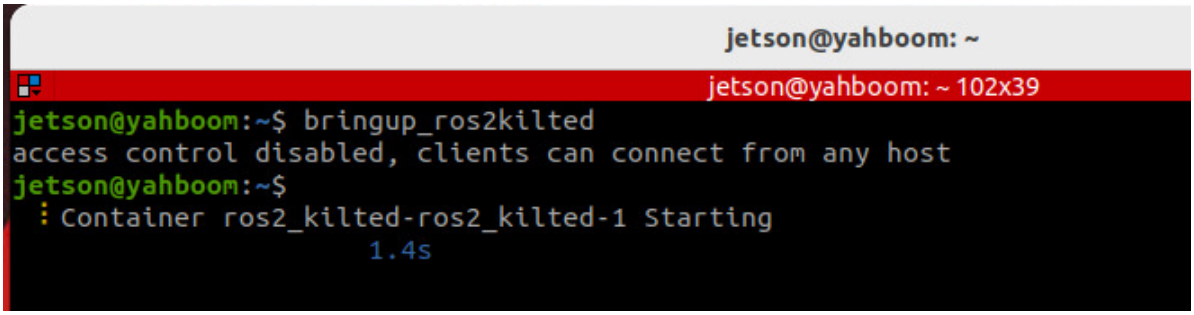
> [!WARNING]
>
> If you use the `start_agent` command to start the microROS agent, you need to close the previous agent first. Note that this is different from the previous microROS agent startup command

```
start_kilted_agent
```



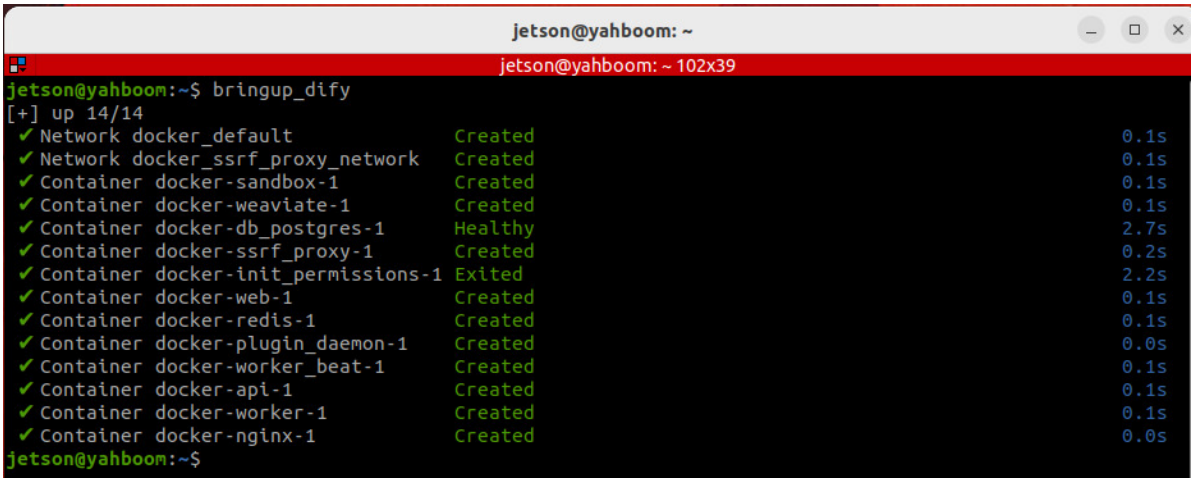
- Start ros2kilted container (if already started, no need to repeat)

```
bringup_ros2kilted
```



- Start Dify, all functions related to AI large models need to start Dify (if already started, no need to repeat)

```
bringup_dify
```



> [!WARNING]
>
> **For Raspberry Pi 5 users, if the ROS container is not started, you need to start the container first,**

```
bringup_m3
```

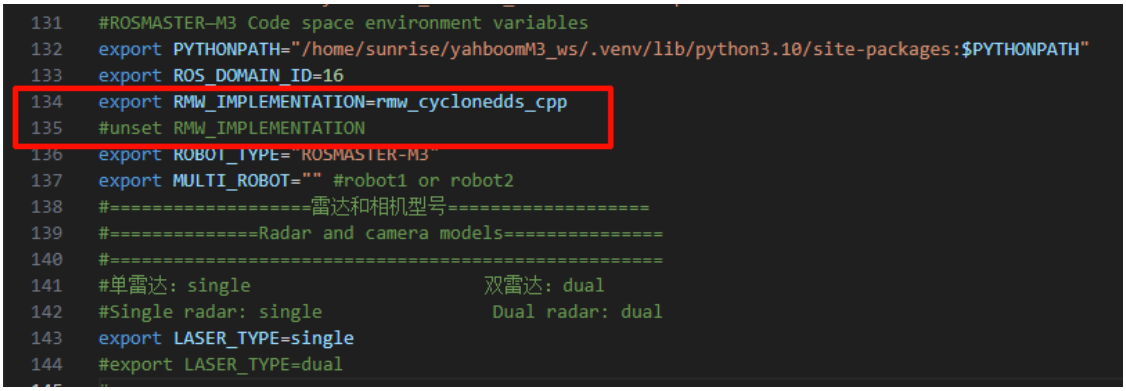
> **Open a container terminal,**

```
exec_m3
```

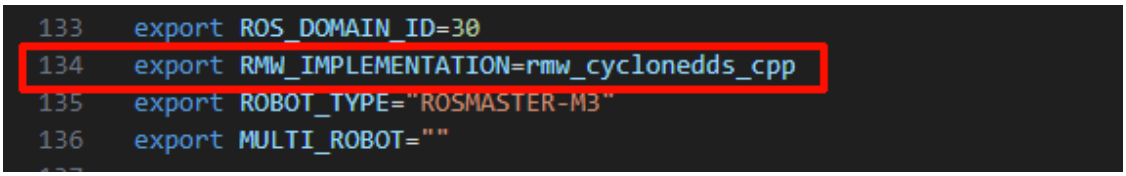
> Open ~/.bashrc, uncomment this line, and change it to the DDS communication middleware as shown in the figure (only the road network planning navigation chapter needs to use cyclondds, **please restore the environment variables after completing the road network chapter!!**)

```
vim ~/.bashrc
```

- RDK & PI5 mainboard



- ORIN mainboard




```
:wq #Save and exit
source ~/.bashrc #Update environment
```

2.2 Prerequisite Knowledge

For a better experience, please first learn and master the following content in the product tutorial:

- AI Large Model: 4. AI Large Model Development, [5.AI Large Model-Voice Version—03-Visual Understanding+SLAM Navigation]
- Road Network Planning Navigation: 7. Road Network Planning Navigation

2.3 Configure Map Mapping File for Road Network Navigation Mode

- For methods to obtain vehicle target points on the map, please refer to [05.AI Model-Voice Version]—[3.Multimodal visual understanding+SLAM navigation—2.3 Configure Map Mapping File]. The method is completely the same, with the difference being the fields filled in the `map_mapping.yaml` map mapping file
- Edit the map mapping file (any editing method can be used)

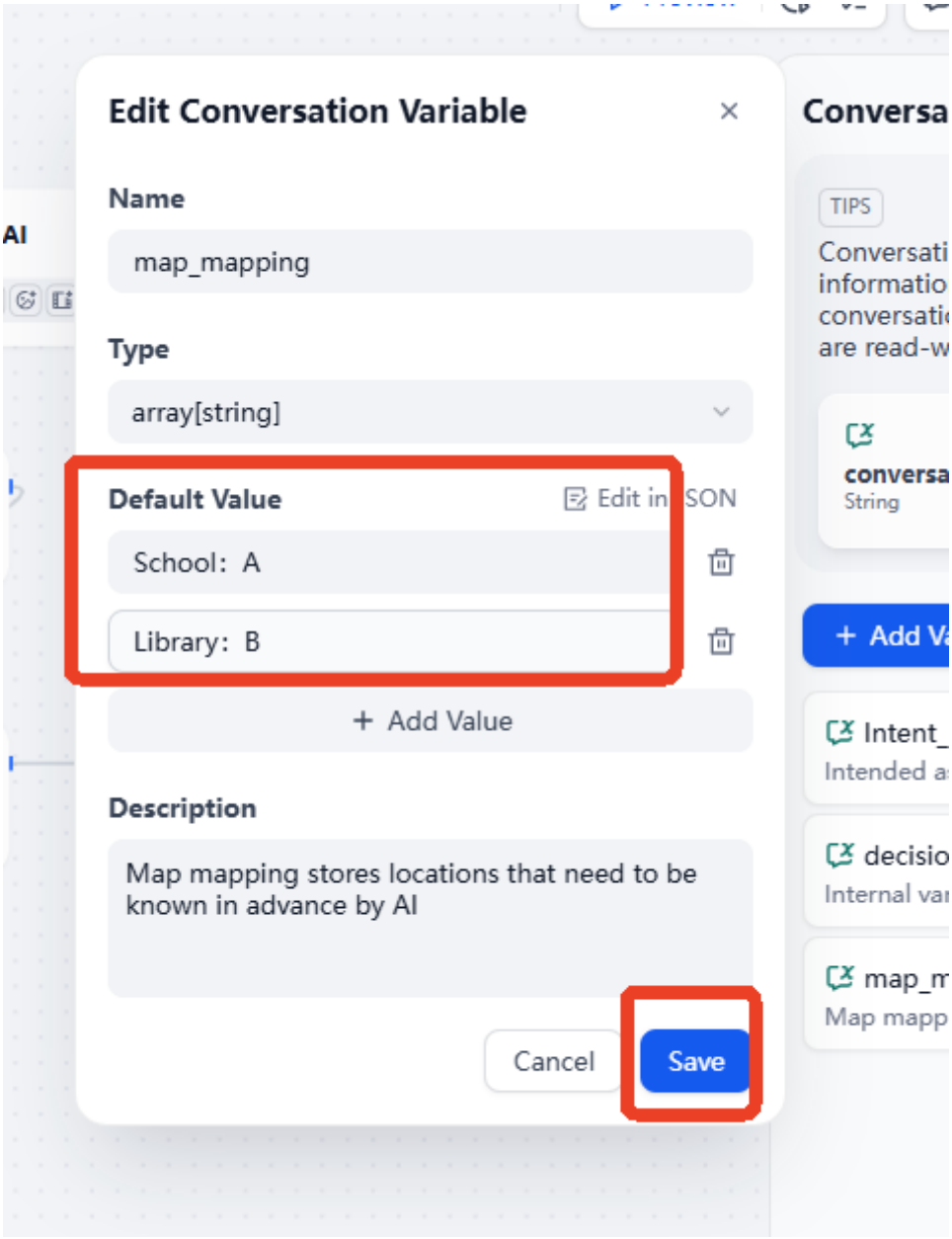
```
~/yahboomM3_ws/user_ws/src/m3_bringup/config/multi_brains/map_mapping.yaml
```

- Road network points need to be filled into the points under the `road_net_map_areas` field. Add or delete according to your needs

```yaml
road_net_map_areas: #Road network navigation
  C:
    name: 'School'
    position:
      x: 2.625
      y: -1.416
      z: 0.0
    orientation:
      x: 0.0
      y: 0.0
      z: -0.990
      w:  0.139

  D:
    name: 'Safe Area'
    position:
      x: 1.583
      y: -0.455
      z: 0.0
    orientation:
      x: 0.0
      y: 0.0
      z: -1.000
      w: -0.007
  E:
    name: 'xxx'
    position:
      x: 0.881
      y: -1.369
      z: 0.0
    orientation:
      x: 0.0
      y: 0.0
      z: -0.004
      w: 1.000
```

- Additionally, you need to update the new map mapping information in Dify and publish save settings. For configuration methods, refer to [05.AI Model-Voice Version]—[3.Multimodal visual understanding+SLAM navigation—2.4 Configure Map Mapping Variables in Dify]



3. Run Case Program

> [!IMPORTANT]
>
> - Need to build the track map's grid map: map, pose map: pose_map, and road network description file: roadnet_file in advance according to [08.Autonomous Driving expand(Need purchase track map)]—[13.Road Network Planning with Track Map]

3.1 Start Road Network Navigation

- Open a terminal on the host machine to start the vehicle's radar fusion node and odometer fusion
- (PI5 users start this command in the `m3 container`)

```
ros2 launch m3_bringup car_base.launch.py
```

- Start ros2_kilted container

```
bringup_ros2kilted
```

- Open ros2kilted container terminal

```
exec_ros2kilted
```

- (RDK & PI5 mainboard) Visualization is started on the accompanying virtual machine `Rosmaster-RoadNet-Ubuntu24.04`,

```
rviz2 -d /home/yahboom/code_ws/src/road_net_route/rviz/nav2_view.rviz
```

- Start road network planning navigation node in container terminal

```
ros2 launch road_net_route roadnet_nav2.launch.py
```

> [!NOTE]
>
> **Parameter Description**
>
> **map** Grid map file name to be loaded, default value is `map.yaml`
>
> **pose_map** Pose map file to be loaded. When navigation global positioning uses slam_toolbox pure positioning, the pose map must be passed in, default value is `map`
>
> **roadnet_file** Road network description file, default value is `map.geojson`

3.2 Start multi_brains Agent

For Raspberry Pi 5 users, if the ROS container is not started, you need to start the container first,

```
bringup_m3
```

Open a container terminal,

```
exec_m3
```

Open a terminal on the vehicle and enter the command to start:

```
ros2 launch m3_bringup llm_agent_control.launch.py enable_route_nav:=True
```

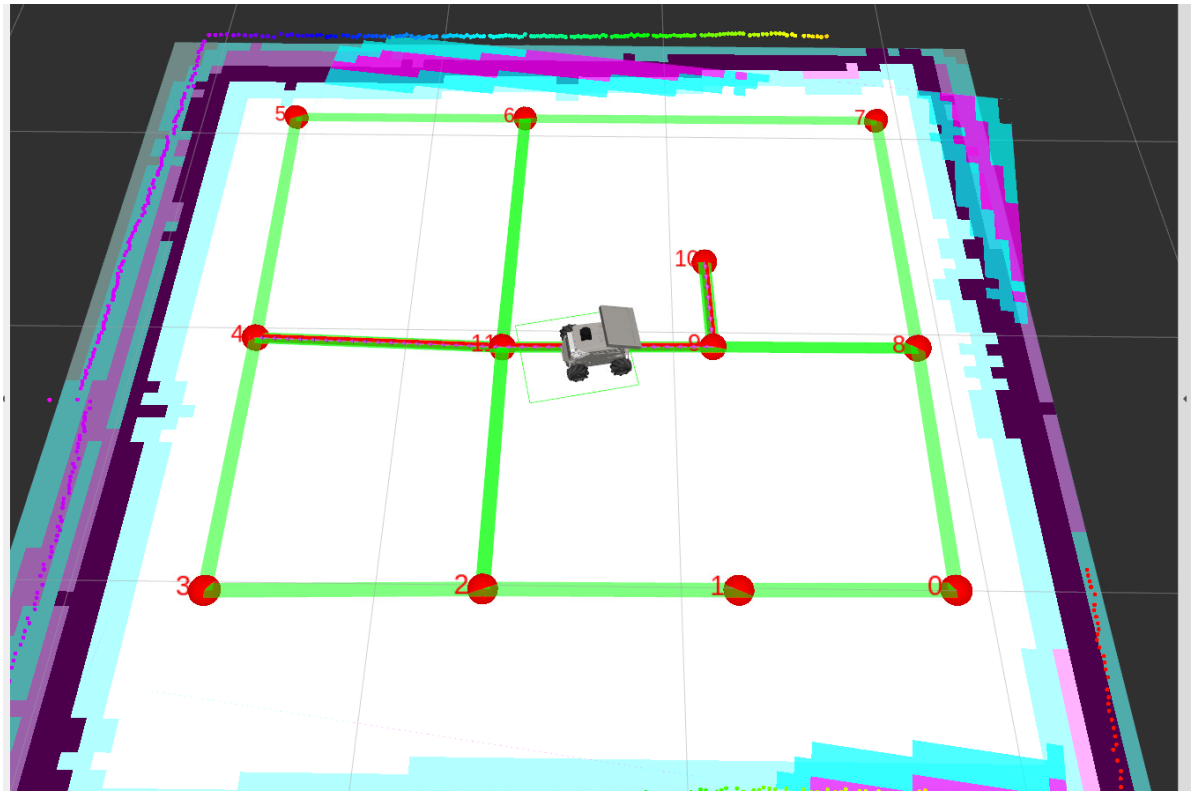
```
[model_service-12] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.modem
[model_service-12] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.modem
[model_service-12] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.phoneline
[model_service-12] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.phoneline
[model_service-12] Cannot connect to server socket err = No such file or directory
[INFO] [model_service-12]: process started with pid [116868]
[INFO] [action_service_usb-13]: process started with pid [116870]
[action_service_usb-13] [INFO] [1766401793.044243388] [action_service_usb]: enable_route_nav: True
[action_service_usb-13] [INFO] [1766401793.152994034] [action_service_usb]: ROS_Action_Service_Initialization completed
[model_service-12] [INFO] [1766401802.719665243] [model_service]: Dify LLM connection successful
[model_service-12] [WARN] [1766401802.720780204] [rcl.logging_rosout]: Publisher already registered for provided node name. If this is due to multipl
e nodes with the same name then all logs for that logger name will go out over the existing publisher. As soon as any node with that name is destruct
ed it will unregister the publisher, preventing any further logs for that name from being published on the rosout topic.
[model_service-12] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.rear
[model_service-12] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.center_lfe
[model_service-12] ALSA lib pcm_oss.c:397:(_snd_pcm_oss_open) Cannot open device /dev/dsp
[model_service-12] ALSA lib confmisc.c:160:(snd_config_get_card) Invalid field card
[model_service-12] ALSA lib pcm_usb_stream.c:482:(_snd_pcm_usb_stream_open) Invalid card 'card'
[model_service-12] ALSA lib confmisc.c:160:(snd_config_get_card) Invalid field card
[model_service-12] ALSA lib pcm_usb_stream.c:482:(_snd_pcm_usb_stream_open) Invalid card 'card'
[model_service-12] ALSA lib pcm_dmix.c:1005:(snd_pcm_dmix_open) The dmix plugin supports only playback stream
[model_service-12] Cannot connect to server socket err = No such file or directory
[model_service-12] Cannot connect to server request channel
[model_service-12] jack server is not running or cannot be started
[model_service-12] JackShmReadWritePtr::~JackShmReadWritePtr - Init not done for -1, skipping unlock
[model_service-12] JackShmReadWritePtr::~JackShmReadWritePtr - Init not done for -1, skipping unlock
[model_service-12] [INFO] [1766401803.247590096] [ASR_Detect]: paraformer-realtime-v2 ASR Engine initialized successfully
[model_service-12] [INFO] [1766401803.262672610] [model_service]: Tongyi TTS Engine initialized successfully
[model_service-12] [INFO] [1766401803.263389574] [model_service]: ROS_LLM_Service Initialization completed
```

3.3 Demonstration Case

- Navigate to the school and observe the surrounding environment

After waking up the vehicle with "Hello, yahboom", when the vehicle responds and the recording prompt sound "dang—" appears, give the command to the vehicle

- The vehicle will then follow the road network planned route to the target point "School" and observe the environment. The route will only travel within the road network



4. Source Code Analysis

Source code path:

```
~/yahboomM3_ws/code_ws/src/multi_brains/multi_brains/action_service.py
```

- In the action_service node initialization, the load_target_points method will be called to load the map_mapping.yaml map mapping file, and the content of the road_net_map_areas field in the configuration file will be stored in the self.road_net_dict variable.

```
    def load_target_points(self):
        """
        Load map mapping file
        """
        self.navpose_dict = {}
        self.road_net_dict = {}
        with open(self.map_mapping_file, "r") as file:
            full_target_points = yaml.safe_load(file)
            common_target_points = full_target_points.get("common_map_areas",
{})
            road_net_target_points =
full_target_points.get("road_net_map_areas", {})

----------------------------------------

        for name, data in road_net_target_points.items():#Load road network
navigation points
            pose = PoseStamped()
            pose.header.frame_id = "map"
            pose.pose.position.x = data["position"]["x"]
            pose.pose.position.y = data["position"]["y"]
            pose.pose.position.z = data["position"]["z"]
            pose.pose.orientation.x = data["orientation"]["x"]
            pose.pose.orientation.y = data["orientation"]["y"]
            pose.pose.orientation.z = data["orientation"]["z"]
            pose.pose.orientation.w = data["orientation"]["w"]
            self.road_net_dict[name] = pose
```

- When the AI large model calls the navigation method, it will judge based on the startup parameter enable_route_nav. If road network navigation mode is started, it will use the __road_net_navigation method for navigation
- __road_net_navigation will construct target pose information and send road network navigation target requests to the route_bridge node through the road_net_nav_pub topic publisher for road network navigation

```
    def navigation(self, point_name):
        """
        Get target point coordinates from navpose_dict dictionary and navigate to
the target point
        """
        if self.enable_route_nav:
            # Use road network navigation
            if self.__road_net_navigation(point_name):
                return True
            else:
                return None
        else:
            # Use normal navigation
            if self.__normal_navigation(point_name):
                return True
```

```
            else:
                return None

    def __road_net_navigation(self, point_name):
        '''Road network navigation function'''
        self.road_net_nav_future = Future()
        goal_msg=PoseStamped()
        #Construct endpoint
        point_name = point_name.strip("'\"")
        goal_msg = self.road_net_dict.get(point_name)
        self.road_net_nav_pub.publish(goal_msg)

        while not self.road_net_nav_future.done():
            if self.interrupt_event.is_set():
                self.cancel_nav_pub.publish(Bool(data=True))
                self.get_logger().info(Fore.GREEN+"Road net navigation
None"+Fore.RESET)
            time.sleep(0.1)
        result = self.road_net_nav_future.result()
        self.__disable_alarm()
        if result.data =="road_net_nav_succeeded":
            self.get_logger().info(Fore.GREEN+"Road net navigation
succeeded"+Fore.RESET)
            return True
        else:
            return False
```